

KOCKNET EMMANUELLA

Machine Learning Engineer

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Professional Summary

Machine Learning Engineer experienced in building end-to-end AI systems — spanning machine learning, deep learning, computer vision, NLP, and Edge AI — from data preprocessing and model development through to deployment-oriented applications using Python and modern AI frameworks. Skilled at applying software engineering principles to build scalable, production-oriented ML solutions, including computer vision models, speech-enabled systems, and retrieval-augmented applications.

Professional Experience

- Intern — Machine Learning & Embedded Systems,**03/2025 – 10/2025
*National Centre for Artificial Intelligence and Robotics (NCAIR)*Abuja
 - Built and evaluated machine learning models using Python, Scikit-learn, and TensorFlow, covering data preprocessing, feature engineering, and exploratory data analysis
 - Applied software engineering principles to develop AI-based solutions for real-world engineering problems
 - Gained hands-on experience integrating software with embedded hardware platforms and microcontrollers
- Intern — SIWES I (Artificial Intelligence & Data Science),**07/2024 – 09/2024
*National Centre for Artificial Intelligence and Robotics (NCAIR)*Abuja
 - Introduced to core concepts in Artificial Intelligence, Data Science, and Python programming through structured, hands-on training
 - Collaborated on team projects including an automatic spending tracker, a document-to-speech converter, and an emotion detection system that played music based on the detected emotion

Education

- Bachelor of Engineering (B.Eng.) in Mechatronics Engineering,**2026
Air Force Institute of Technology (AFIT), Kaduna, Nigeria

Technical Skills

- Machine Learning:**
Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Cross-Validation, Ensemble Learning, Time-Series Modeling, Scikit-learn, XGBoost, LightGBM, CatBoost
- MLOps/Deployment:**
Model Deployment, REST APIs, FastAPI, Docker, CI/CD, ML Pipelines
- Data:**
Data Preprocessing, Data Cleaning, Exploratory Data Analysis (EDA), SQL, Pandas, NumPy, Matplotlib, Data Visualization, Feature Engineering

FEATURED PROJECTS

Predictive Maintenance System [↗](#), *Aircraft Engine RUL Prediction (End-to-End ML Product)*

Tech Stack: Python, CatBoost, LSTM, SHAP, FastAPI, JWT, Flutter, SQLAlchemy Built an end-to-end ML system predicting aircraft engine Remaining Useful Life (RUL) using NASA's C-MAPSS dataset — from model development to a deployed mobile application.

- Benchmarked 5 models (Random Forest, XGBoost, LightGBM, CatBoost, LSTM); achieved MAE 10.37 / RMSE 14.23 with LSTM, selected CatBoost (RMSE 16.70) for production based on inference simplicity and SHAP explainability
- Built a FastAPI backend with JWT authentication, bcrypt hashing, and RESTful prediction endpoints (SQLAlchemy/SQLite)
- Developed a native Flutter mobile app with real-time predictions, interactive sensor/RUL charts, and persistent authenticated sessions, tested on physical Android hardware

Mechatronics Tutor with Edge AI [↗](#)

Tech Stack: Python, Edge AI, ChromaDB, RAG, Speech-to-Text, Text-to-Speech, Raspberry Pi Offline Edge AI tutoring system for mechatronics students, combining semantic search, speech recognition, and speech synthesis on resource-constrained hardware.

Fruit Classification Using CNNs [↗](#)

Tech Stack: Python, TensorFlow, Keras, CNN, Computer Vision Image classification model identifying fruit categories, using data augmentation and deep learning to improve generalization.

AI Model Trends Analysis [↗](#)

Tech Stack: Python, Pandas, NumPy, Matplotlib, EDA Exploratory analysis of Epoch AI's Notable AI Models dataset, examining growth in training compute, parameters, and top contributing organizations.

Student Performance Analysis [↗](#)

Tech Stack: Python, Pandas, Data Analysis, Visualization Exploratory analysis of study habits and academic performance, identifying patterns that drive student outcomes.

Multilingual Document-to-Speech Converter [↗](#)

Tech Stack: Python, NLP, Speech-to-Text, Text-to-Speech, GUI Development Desktop application translating documents into multiple languages and generating natural speech output via a graphical interface.

Certifications

IBM Data Science Professional Certificate

Issuer: IBM

Supervised Machine Learning

Issuer: DeepLearning.AI

AI Fluency

Issuer: Anthropic

Advanced Python Programming & Embedded Systems

Issuer: NCAIR

Introduction to SQL

Issuer: Kaggle